



# Exploring the potential of pre-trained CNN models for robust maternal–fetal ultrasound plane classification

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## ABSTRACT

Maternal–fetal ultrasound imaging plays an important role in prenatal care, facilitating the assessment of fetal development and maternal health. However, accurately identifying standard fetal ultrasound planes remains challenging due to subjective interpretation. This paper presents a method that explores pre-trained convolutional neural networks (CNNs) trained on the IMAGENET database to achieve effective classification without requiring specific fine-tuning of ultrasound data. In this work, lightweight pre-trained CNNs such as MobileNetV2, SqueezeNet, and ShuffleNet are chosen. Features extracted from the final dense layers of pre-trained CNNs are fed into individual support vector machines (SVMs), each trained for a specific speckle noise levels. The classification decisions from these SVMs are then combined using majority voting to improve overall performance. To address speckle noise, specific SVMs are trained for different levels of speckle noise. For obtaining consistent performance, the noise level in the input ultrasound image is mapped to the corresponding SVM trained for that noise level using a proposed switch control mechanism. The presented approach demonstrates promising results in identifying critical standard ultrasound plane structures such as the fetal abdomen, brain, cervix, femur, thorax, and non-standard planes. Through extensive experimentation using 12,400 ultrasound images from the open-access FETAL\_PLANES\_DB dataset, the potential of pre-trained CNNs in maternal–fetal ultrasound plane classification is demonstrated. The proposed method eliminates the challenges associated with fine-tuning CNNs, which can be resource-intensive and computationally demanding. From the experimental analysis, the proposed method achieves a diagnostic accuracy of 90.17%, comparable to various fine-tuned CNN based approaches reported in the literature.

## 1. Introduction

The identification of reliable standard planes of fetal anatomy is a crucial step in clinical obstetric examinations for an accurate diagnosis of fetal development [1]. Obstetricians rely solely on their experience to select the appropriate standard plane, ensuring that essential fetal structures are visualized and measured accurately [2]. The manual process of selecting the standard fetal ultrasound plane allows obstetricians to evaluate various structural parameters to provide valuable insights into fetal growth and development. The identification of standard planes is essential for the accurate assessment of fetal growth parameters, such as head circumference, abdominal circumference, and femur length [3]. However, manual acquisition of reliable fetal ultrasound planes is labor-intensive and time-consuming [4]. Therefore, automated techniques for identifying standard fetal ultrasound biometry planes can streamline scanning and improve detection efficiency.

Standard ultrasound anatomical plane structures such as fetal abdomen, brain, cervix, femur, thorax, and non-standard planes, constitute crucial focal points in maternal–fetal ultrasound examinations as shown in Fig. 1. Accurate identification of maternal–fetal ultrasound planes are critical in prenatal care [6]. In literature, the use of convolutional neural networks (CNNs) enables automated recognition of standard fetal planes with remarkable accuracy, significantly reducing the burden on sonographers [7]. By utilizing large datasets of annotated ultrasound images, the fine-tuned CNNs can learn complex patterns, enhancing the efficiency of fetal plane recognition. But the following concerns are observed in the use of fine-tuned CNN architectures in the maternal–fetal plane classification.

- Access to large and diverse annotated datasets for training deep learning models remains a challenge.

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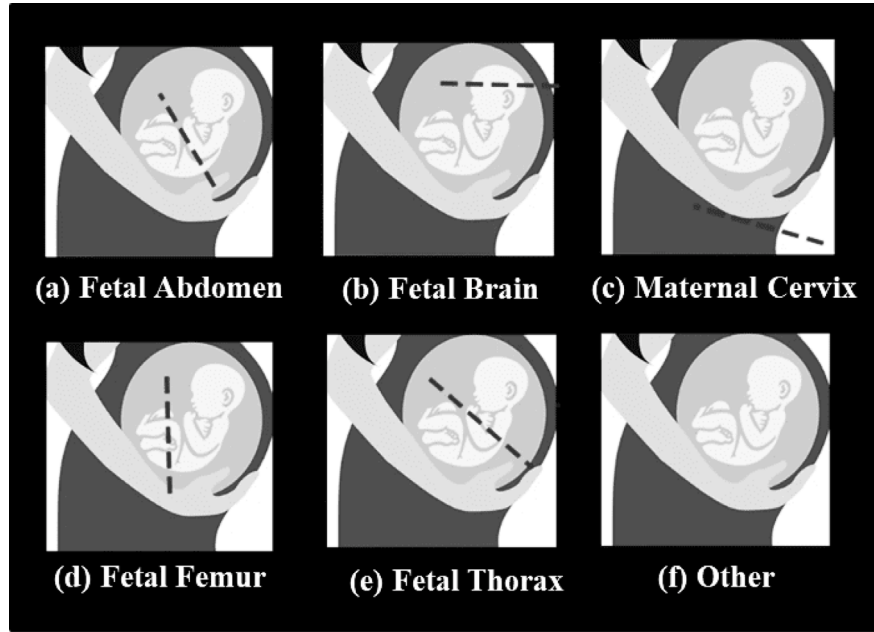


Fig. 1. Maternal-fetal anatomical planes: (a) fetal abdomen, (b) fetal brain, (c) maternal cervix, (d) fetal femur, (e) fetal thorax, and (f) other non-standard planes [5].

- The fine-tuning of CNN for a particular maternal-fetal ultrasound dataset is computationally expensive.
- Models trained on datasets from high-resource settings may struggle to generalize to populations or clinical settings with different demographics or imaging protocols.

In addition to these concerns, it is evident that speckle noise often deteriorates the performance of fetal ultrasound plane classification. Ultrasound image quality is frequently compromised by speckle noise, an inherent artifact resulting from the coherent nature of ultrasound waves [8]. The speckle noise obscures fine details and adversely affects the accuracy of feature extraction and diagnostic measurements. Addressing speckle noise is crucial for ensuring reliable analysis in fetal ultrasound, where precise anatomical identification and measurements are essential [9]. Therefore, in the proposed method, speckle noise is systematically analyzed to improve the robustness of the classification process.

Motivated by the preceding concerns, this work proposes a method that utilizes pre-trained CNNs for feature extraction, followed by classification with traditional methods like support vector machine (SVM). The proposed approach for maternal-fetal ultrasound plane classification offers several advantages compared to fine-tuning the entire CNN as follows:

- Pre-trained CNNs have learned generic features from large-scale datasets, facilitating their transferability across different tasks and domains.
- Utilizing pre-trained CNNs for feature extraction reduces the need for extensive annotated data, making it more feasible in scenarios with limited labeled data.
- Feature extraction with pre-trained CNNs is computationally more efficient compared to fine-tuning the entire network.
- The proposed method using a switch control mechanism provides consistent performance even under varying speckle noise conditions, ensuring robustness.

The presented methodology provides a flexible and efficient framework for fetal plane classification in obstetric ultrasound imaging, addressing key challenges such as variability and data availability.

The remainder of this paper is organized as follows. Section 2 reviews related work on the classification of the fetal plane. Section 3 outlines the methodology, including feature extraction with pre-trained networks and SVM classification under noisy conditions. In Section 4, experimental results and performance evaluations are presented, followed by analysis. Section 5 provides a detailed discussion of the findings and their implications. Finally, Section 6 concludes the paper, emphasizing its contribution to fetal ultrasound imaging and diagnosis.

## 2. Related works

The precision of fetal measurements during ultrasound is essential to scrutinize machine calibration, measurement chart accuracy, and image quality to ensure accurate clinical decisions [10]. The research on fetal plane localization and classification has intensified to minimize measurement errors. However, the subjective nature of fetal plane localization emphasize the urgent need for automated assessment techniques to enhance measurement accuracy and refine clinical interventions [5].

In recent years, researchers have explored various algorithms for maternal-fetal plane classification and localization. The challenge involves identifying specific category from the input ultrasound image, such as the fetal abdomen, brain, cervix, femur, thorax, and others, as illustrated in Fig. 2. The collection of datasets is essential for validating maternal-fetal ultrasound plane classification. These datasets provide a foundation for training and testing CNN models under diverse conditions, ensuring robust and reliable classification. Comprehensive information of datasets related to maternal-fetal ultrasound plane classification, including their accessibility, size, and use cases, is available in Ref. [6]. The authors in [11] proposed an algorithm utilizing dense sampling feature transform and SVM to automatically identify fetal planes in ultrasound images with high accuracy across various gestational ages. In [12], a domain-transferred deep CNN approach, trained on 11,942 expert-annotated fetal abdominal ultrasound images, enhanced fetal abdominal standard plane localization in ultrasound videos. The presented approach achieving better performance through transfer learning. A CNN-based method presented in [13] enables automatic detection of 13 fetal standard views and localizing fetal structures

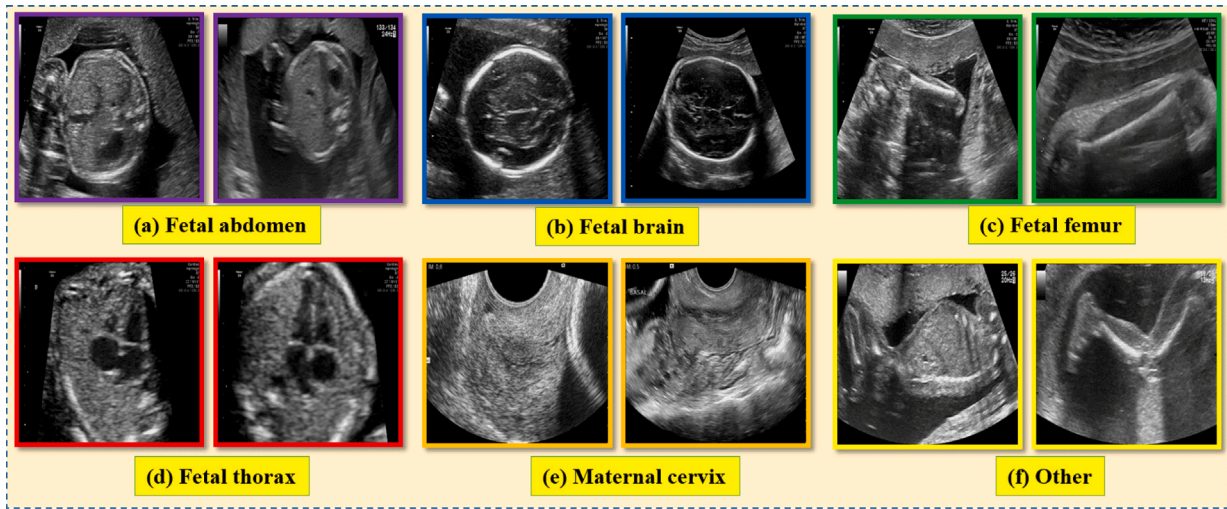


Fig. 2. Maternal-fetal ultrasound planes: (a) fetal abdomen, (b) fetal brain, (c) fetal femur, (d) fetal thorax, (e) maternal cervix, and (f) other [5].

in 2-D ultrasound data, demonstrating real-time performance and high accuracy. Unlike prior studies focusing on individual planes, the approach in [14] offers a general solution for identifying various standard planes, demonstrating improved efficacy through multitask learning on a large dataset. To improve fetal facial standard plane recognition, a CNN with 16 convolutional layers, global average pooling, and transfer learning is proposed [15]. SPRNet [16], which utilizes the DenseNet architecture and data-driven partial transfer learning, achieves 99% accuracy on their private dataset for the automatic recognition of fetal standard planes in ultrasound images. A novel approach in [17] has been used differential-CNN for identifying fetal brain ultrasound standard planes, achieves 92.93% accuracy without added complexity. Exploiting the Industrial Internet of Things framework, authors in [18] developed an automatic fetal ultrasound standard plane recognition model, combining CNN and recurrent neural networks to achieve superior accuracy and robustness in identifying fetal structures across ultrasound frames. The authors in [19] proposed an automated framework based on a 2D+t CNN, enhancing first-trimester ultrasound assessment and providing insights into operator skill and variability.

Open-access datasets are essential for conducting extensive evaluations and ensuring robust performance of classification models. The availability of large open-access datasets plays a crucial role in advancing research in maternal-fetal ultrasound imaging. In [5], the authors released a large open-access dataset named FETAL\_PLANES\_DB,<sup>1</sup> containing 12,400 ultrasound fetal planes. This dataset provides a significant resource for developing and benchmarking algorithms in fetal plane classification. In [5], the authors assessed deep learning classification techniques using FETAL\_PLANES\_DB [5] dataset, showcasing comparable performance to human technicians in classifying fetal anatomical planes. A deep-learning model for automating fetal organ classification from ultrasound images is proposed in [7], with the potential to improve maternal health services worldwide. A method integrating deep features from AlexNet and VGG-19, using a global average pooling layer followed by a multi-layer perceptron, effectively classifies ultrasound images into six categories with improved accuracy and efficiency metrics [20]. Another approach utilizes a pre-trained VGG-19 model combined with adaptive channel weighting and multiscale channel attention modules to recalibrate feature map channels and address low inter-class variance [21]. Furthermore, the VGGSA network incorporates a spatial attention module

into a pre-trained VGG-19 architecture to enhance feature representation, mitigating misclassifications caused by anatomical structural similarities [22].

These studies collectively highlight the advancements in utilizing CNNs for the automatic classification of maternal fetal ultrasound planes. The research aims to enhance the efficiency and accuracy of prenatal diagnosis by automating the recognition of standard imaging planes, which is crucial for monitoring fetal development and detecting abnormalities.

### 3. Methodology

In this section, the power of a pre-trained CNN model is exploited for feature extraction from maternal-fetal ultrasound images. By utilizing a CNN pre-trained on the IMAGENET [23] database, it serves as a powerful feature extractor capable of capturing important spatial and structural information relevant to ultrasound imaging. The extracted features retain valuable information about fetal development and maternal health, encapsulated in high-dimensional representations. Subsequently, these feature vectors are fed into an SVM classifier for accurate classification of ultrasound fetal planes. By combining the discriminative power of SVM with the comprehensive feature representations obtained from the pre-trained CNN, robust and effective classification performance is achieved. The framework of the proposed methodology is presented in Fig. 3.

Let  $X$  be the set of maternal-fetal ultrasound images, where each image  $x_i \in X$  is represented as a matrix of pixel values. Formally, we define  $x_i$  as a two-dimensional color maternal-fetal ultrasound plane of dimensions  $W \times H$ , where  $W$  represents the width and  $H$  represents the height of the image. Each pixel in the matrix is characterized by its intensity value. The objective of the classification task is to assign each ultrasound image  $x_i$  to one of the standard ultrasound imaging planes denoted by the set  $Y = \{y_1, y_2, \dots, y_n\}$ . Each  $y_j \in Y$  represents a specific standard imaging plane, such as the abdomen, brain, cervix, femur, thorax, and other. Thus, the classification problem can be formulated as a mapping function  $f : X \rightarrow Y$ , which assigns each image  $x_i$  to its corresponding  $y_j$ . Mathematically, the classification process involves learning a discriminative function  $f$  that maps the input space  $X$  to the output space  $Y$ . This function is typically learned from a labeled dataset  $\{(x_i, y_i)\}_{i=1}^N$ , where  $N$  is the number of training samples.

#### 3.1. Speckle noise model

Let  $x_{n,i}$  represent the ultrasound image with speckle noise, which can be modeled as the sum of the original ultrasound image  $x_i$  and

<sup>1</sup> <https://zenodo.org/records/3904280>.

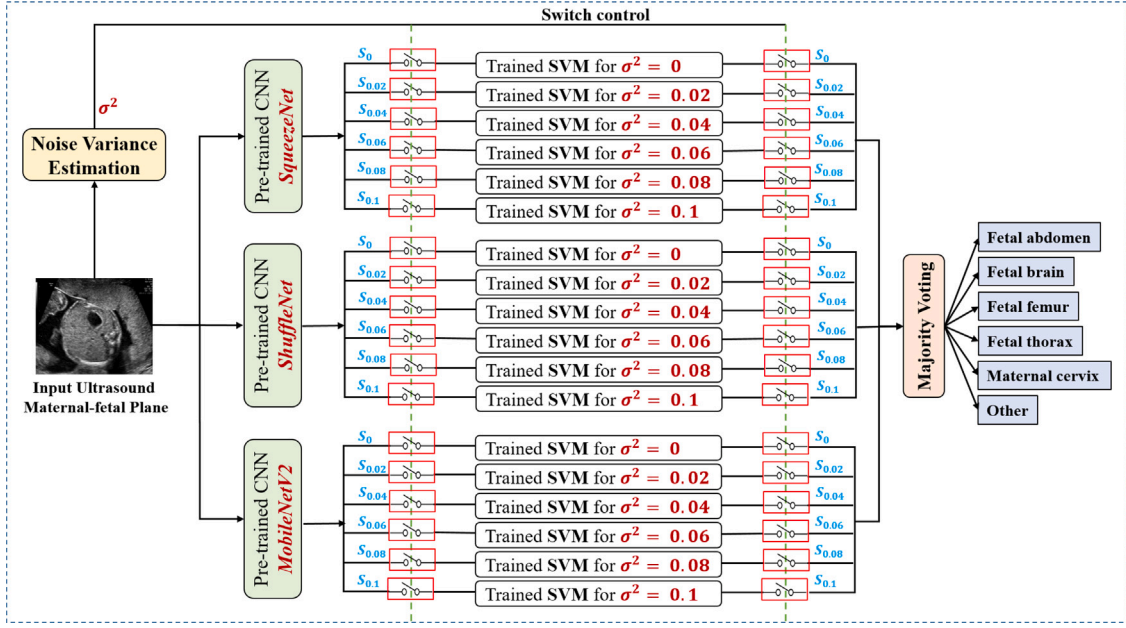


Fig. 3. Framework for maternal-fetal ultrasound plane classification under speckle noise conditions.

|              |   | $\sigma^2 = 0$    |               |                |                |               |                |                |
|--------------|---|-------------------|---------------|----------------|----------------|---------------|----------------|----------------|
| OUTPUT CLASS | A | 274<br>5.2%       | 3<br>0.1%     | 1<br>0.0%      | 11<br>0.2%     | 1<br>0.0%     | 46<br>0.9%     | 81.5%<br>18.5% |
|              | B | 8<br>0.2%         | 1461<br>27.7% | 3<br>0.1%      | 2<br>0.0%      | 0<br>0.0%     | 29<br>0.6%     | 97.2%<br>2.8%  |
|              | C | 3<br>0.1%         | 0<br>0.0%     | 419<br>7.9%    | 3<br>0.1%      | 0<br>0.0%     | 99<br>1.9%     | 80.0%<br>20.0% |
|              | D | 25<br>0.5%        | 3<br>0.1%     | 0<br>0.0%      | 592<br>11.2%   | 0<br>0.0%     | 70<br>1.3%     | 85.8%<br>14.2% |
|              | E | 0<br>0.0%         | 3<br>0.1%     | 0<br>0.0%      | 0<br>0.0%      | 643<br>12.2%  | 4<br>0.1%      | 98.9%<br>1.1%  |
|              | F | 48<br>0.9%        | 2<br>0.0%     | 101<br>2.0%    | 52<br>1.0%     | 1<br>0.0%     | 1364<br>25.9%  | 87.0%<br>13.0% |
|              |   | 76.5%<br>23.5%    | 99.3%<br>0.7% | 80.0%<br>20.0% | 89.7%<br>10.3% | 99.7%<br>0.3% | 84.6%<br>15.4% | 90.2%<br>9.8%  |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.02$ |               |                |                |               |                |                |
|              | A | 273<br>5.2%       | 4<br>0.1%     | 2<br>0.0%      | 13<br>0.2%     | 1<br>0.0%     | 53<br>1.0%     | 78.9%<br>21.1% |
|              | B | 6<br>0.1%         | 1459<br>27.7% | 4<br>0.1%      | 4<br>0.0%      | 0<br>0.0%     | 32<br>0.6%     | 96.9%<br>3.1%  |
|              | C | 4<br>0.1%         | 0<br>0.0%     | 420<br>8.0%    | 0<br>0.0%      | 0<br>0.0%     | 91<br>1.7%     | 81.6%<br>18.4% |
|              | D | 22<br>0.4%        | 2<br>0.0%     | 0<br>0.0%      | 585<br>11.1%   | 0<br>0.0%     | 78<br>1.5%     | 85.2%<br>14.8% |
|              | E | 0<br>0.0%         | 2<br>0.0%     | 0<br>0.0%      | 0<br>0.0%      | 644<br>12.2%  | 4<br>0.1%      | 99.1%<br>0.9%  |
|              | F | 53<br>1.0%        | 5<br>0.1%     | 98<br>1.9%     | 58<br>1.1%     | 0<br>0.0%     | 1354<br>25.7%  | 86.4%<br>13.6% |
|              |   | 76.3%<br>23.7%    | 99.1%<br>0.9% | 80.2%<br>19.8% | 88.6%<br>11.4% | 99.8%<br>0.2% | 84.0%<br>16.0% | 89.8%<br>10.2% |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.04$ |               |                |                |               |                |                |
|              | A | 252<br>4.8%       | 3<br>0.1%     | 3<br>0.1%      | 14<br>0.3%     | 1<br>0.0%     | 53<br>1.0%     | 77.3%<br>22.7% |
|              | B | 5<br>0.1%         | 1458<br>27.7% | 1<br>0.0%      | 4<br>0.1%      | 0<br>0.0%     | 34<br>0.6%     | 97.1%<br>2.9%  |
|              | C | 4<br>0.1%         | 0<br>0.0%     | 410<br>7.8%    | 3<br>0.1%      | 0<br>0.0%     | 77<br>1.5%     | 83.0%<br>17.0% |
|              | D | 20<br>0.4%        | 2<br>0.0%     | 2<br>0.0%      | 574<br>10.9%   | 0<br>0.0%     | 70<br>1.3%     | 85.9%<br>14.1% |
|              | E | 0<br>0.0%         | 3<br>0.1%     | 0<br>0.0%      | 0<br>0.0%      | 644<br>12.2%  | 4<br>0.1%      | 98.9%<br>1.1%  |
|              | F | 77<br>1.5%        | 6<br>0.1%     | 108<br>2.0%    | 65<br>1.3%     | 0<br>0.0%     | 1374<br>26.1%  | 84.3%<br>15.7% |
|              |   | 70.4%<br>29.6%    | 99.0%<br>1.0% | 78.2%<br>21.8% | 87.0%<br>13.0% | 99.8%<br>0.2% | 85.2%<br>14.8% | 89.4%<br>10.6% |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.06$ |               |                |                |               |                |                |
|              | A | 255<br>4.8%       | 3<br>0.1%     | 1<br>0.0%      | 14<br>0.3%     | 1<br>0.0%     | 56<br>1.1%     | 77.3%<br>22.7% |
|              | B | 5<br>0.1%         | 1454<br>27.6% | 6<br>0.1%      | 1<br>0.0%      | 0<br>0.0%     | 28<br>0.5%     | 97.3%<br>2.7%  |
|              | C | 6<br>0.1%         | 0<br>0.0%     | 419<br>7.9%    | 5<br>0.1%      | 0<br>0.0%     | 75<br>1.4%     | 83.0%<br>17.0% |
|              | D | 17<br>0.3%        | 2<br>0.0%     | 0<br>0.0%      | 568<br>10.8%   | 0<br>0.0%     | 79<br>1.5%     | 85.3%<br>14.7% |
|              | E | 0<br>0.0%         | 2<br>0.0%     | 0<br>0.0%      | 0<br>0.0%      | 643<br>12.2%  | 4<br>0.1%      | 99.1%<br>0.9%  |
|              | F | 75<br>1.4%        | 11<br>0.2%    | 98<br>1.9%     | 72<br>1.4%     | 1<br>0.0%     | 1370<br>26.0%  | 84.2%<br>15.8% |
|              |   | 71.2%<br>28.8%    | 98.8%<br>1.2% | 80.0%<br>20.0% | 86.1%<br>13.9% | 99.7%<br>0.3% | 85.0%<br>15.0% | 89.3%<br>10.7% |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.08$ |               |                |                |               |                |                |
|              | A | 246<br>4.7%       | 2<br>0.0%     | 1<br>0.0%      | 14<br>0.3%     | 0<br>0.0%     | 57<br>1.1%     | 76.9%<br>23.1% |
|              | B | 7<br>0.1%         | 1452<br>27.5% | 4<br>0.1%      | 6<br>0.1%      | 1<br>0.0%     | 32<br>0.6%     | 96.7%<br>3.3%  |
|              | C | 7<br>0.1%         | 0<br>0.0%     | 426<br>8.1%    | 1<br>0.0%      | 0<br>0.0%     | 78<br>1.5%     | 83.2%<br>16.8% |
|              | D | 26<br>0.5%        | 4<br>0.1%     | 0<br>0.0%      | 581<br>11.0%   | 0<br>0.0%     | 93<br>1.8%     | 82.5%<br>17.5% |
|              | E | 0<br>0.0%         | 2<br>0.0%     | 0<br>0.0%      | 0<br>0.0%      | 643<br>12.2%  | 4<br>0.1%      | 99.1%<br>0.9%  |
|              | F | 72<br>1.4%        | 12<br>0.2%    | 93<br>1.8%     | 58<br>1.1%     | 1<br>0.0%     | 1348<br>25.6%  | 85.1%<br>14.9% |
|              |   | 68.7%<br>31.3%    | 98.6%<br>1.4% | 81.3%<br>18.7% | 88.0%<br>12.0% | 99.7%<br>0.3% | 83.6%<br>16.4% | 89.1%<br>10.9% |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.1$  |               |                |                |               |                |                |
|              | A | 241<br>4.6%       | 5<br>0.1%     | 2<br>0.0%      | 14<br>0.3%     | 0<br>0.0%     | 51<br>1.0%     | 77.0%<br>23.0% |
|              | B | 7<br>0.1%         | 1447<br>27.5% | 4<br>0.1%      | 5<br>0.1%      | 1<br>0.0%     | 26<br>0.5%     | 97.1%<br>2.9%  |
|              | C | 3<br>0.1%         | 0<br>0.0%     | 414<br>7.9%    | 2<br>0.0%      | 0<br>0.0%     | 72<br>1.4%     | 84.3%<br>15.7% |
|              | D | 23<br>0.4%        | 4<br>0.1%     | 1<br>0.0%      | 577<br>10.9%   | 0<br>0.0%     | 80<br>1.5%     | 84.2%<br>15.8% |
|              | E | 0<br>0.0%         | 1<br>0.0%     | 0<br>0.0%      | 0<br>0.0%      | 643<br>12.2%  | 4<br>0.1%      | 99.2%<br>0.8%  |
|              | F | 84<br>1.6%        | 15<br>0.3%    | 103<br>2.0%    | 62<br>1.2%     | 1<br>0.0%     | 1379<br>26.2%  | 83.9%<br>16.1% |
|              |   | 67.3%<br>32.7%    | 98.3%<br>1.7% | 79.0%<br>21.0% | 87.4%<br>12.6% | 99.7%<br>0.3% | 85.5%<br>14.5% | 89.2%<br>10.8% |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   |                   |               |                |                |               |                |                |
|              |   | TARGET CLASS      |               |                |                |               |                |                |

Fig. 4. Performance of the proposed method without NVE at different speckle noise levels (right column: precision, bottom row: recall) for the following classes: A: Fetal Abdomen, B: Fetal Brain, C: Fetal Femur, D: Fetal Thorax, E: Maternal Cervix, F: Other.

speckle noise with variance  $\sigma^2$ :

$$x_{n,i} = x_i + x_i^\gamma \cdot \sigma \cdot \eta$$

where  $\eta$  follows a standard normal distribution  $\mathcal{N}(0, 1)$  and  $\gamma = 1$  is the speckle noise factor [24].

(1)

|              |   | $\sigma^2 = 0$    |               |                |                |               |                |                |
|--------------|---|-------------------|---------------|----------------|----------------|---------------|----------------|----------------|
| OUTPUT CLASS | A | 274<br>5.2%       | 3<br>0.1%     | 1<br>0.0%      | 11<br>0.2%     | 1<br>0.0%     | 46<br>0.9%     | 81.5%          |
|              | B | 8<br>0.2%         | 1461<br>27.7% | 3<br>0.1%      | 2<br>0.0%      | 0<br>0.0%     | 29<br>0.6%     | 97.2%<br>2.8%  |
|              | C | 3<br>0.1%         | 0<br>0.0%     | 419<br>7.9%    | 3<br>0.1%      | 0<br>0.0%     | 99<br>1.9%     | 80.0%          |
|              | D | 25<br>0.5%        | 3<br>0.1%     | 0<br>0.0%      | 592<br>11.2%   | 0<br>0.0%     | 70<br>1.3%     | 85.8%<br>14.2% |
|              | E | 0<br>0.0%         | 3<br>0.1%     | 0<br>0.0%      | 0<br>0.0%      | 643<br>12.2%  | 4<br>0.1%      | 98.9%<br>1.1%  |
|              | F | 48<br>0.9%        | 2<br>0.0%     | 101<br>1.9%    | 52<br>1.0%     | 1<br>0.0%     | 1364<br>25.9%  | 87.0%<br>13.0% |
|              |   | 76.5%<br>23.5%    | 99.3%<br>0.7% | 80.0%<br>20.0% | 89.7%<br>10.3% | 99.7%<br>0.3% | 84.6%<br>15.4% | 90.2%<br>9.8%  |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.02$ |               |                |                |               |                |                |
|              | A | 273<br>5.2%       | 4<br>0.1%     | 2<br>0.0%      | 13<br>0.2%     | 1<br>0.0%     | 53<br>1.0%     | 78.9%<br>21.1% |
|              | B | 6<br>0.1%         | 1459<br>27.7% | 4<br>0.1%      | 4<br>0.1%      | 0<br>0.0%     | 32<br>0.6%     | 96.9%<br>3.1%  |
|              | C | 4<br>0.1%         | 0<br>0.0%     | 420<br>8.0%    | 0<br>0.0%      | 0<br>0.0%     | 91<br>1.7%     | 81.6%<br>18.4% |
|              | D | 22<br>0.4%        | 2<br>0.0%     | 0<br>0.0%      | 585<br>11.1%   | 0<br>0.0%     | 78<br>1.5%     | 85.2%<br>14.8% |
|              | E | 0<br>0.0%         | 2<br>0.0%     | 0<br>0.0%      | 0<br>0.0%      | 644<br>12.2%  | 4<br>0.1%      | 99.1%<br>0.9%  |
|              | F | 53<br>1.0%        | 5<br>0.1%     | 98<br>1.9%     | 58<br>1.1%     | 0<br>0.0%     | 1354<br>25.7%  | 86.4%<br>13.6% |
|              |   | 76.3%<br>23.7%    | 99.1%<br>0.9% | 80.2%<br>19.8% | 88.6%<br>11.4% | 99.8%<br>0.2% | 84.0%<br>16.0% | 89.8%<br>10.2% |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.04$ |               |                |                |               |                |                |
|              | A | 252<br>4.8%       | 3<br>0.1%     | 3<br>0.1%      | 14<br>0.3%     | 1<br>0.0%     | 53<br>1.0%     | 77.3%<br>22.7% |
|              | B | 5<br>0.1%         | 1458<br>27.7% | 1<br>0.0%      | 4<br>0.1%      | 0<br>0.0%     | 34<br>0.6%     | 97.1%<br>2.9%  |
|              | C | 4<br>0.1%         | 0<br>0.0%     | 410<br>7.8%    | 3<br>0.1%      | 0<br>0.0%     | 77<br>1.5%     | 83.0%<br>17.0% |
|              | D | 20<br>0.4%        | 2<br>0.0%     | 2<br>0.0%      | 574<br>10.9%   | 0<br>0.0%     | 70<br>1.3%     | 85.9%<br>14.1% |
|              | E | 0<br>0.0%         | 3<br>0.1%     | 0<br>0.0%      | 0<br>0.0%      | 644<br>12.2%  | 4<br>0.1%      | 98.9%<br>1.1%  |
|              | F | 77<br>1.5%        | 6<br>0.1%     | 108<br>2.0%    | 65<br>1.2%     | 0<br>0.0%     | 1374<br>26.1%  | 84.3%<br>15.7% |
|              |   | 70.4%<br>29.6%    | 99.0%<br>1.0% | 78.2%<br>21.8% | 87.0%<br>13.0% | 99.8%<br>0.2% | 85.2%<br>14.8% | 89.4%<br>10.6% |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.06$ |               |                |                |               |                |                |
|              | A | 255<br>4.8%       | 3<br>0.1%     | 1<br>0.0%      | 14<br>0.3%     | 1<br>0.0%     | 56<br>1.1%     | 77.3%<br>22.7% |
|              | B | 5<br>0.1%         | 1454<br>27.6% | 6<br>0.1%      | 1<br>0.0%      | 0<br>0.0%     | 28<br>0.5%     | 97.3%<br>2.7%  |
|              | C | 6<br>0.1%         | 0<br>0.0%     | 419<br>7.9%    | 5<br>0.1%      | 0<br>0.0%     | 75<br>1.4%     | 83.0%<br>17.0% |
|              | D | 17<br>0.3%        | 0<br>0.0%     | 0<br>0.0%      | 568<br>10.8%   | 0<br>0.0%     | 79<br>1.5%     | 85.3%<br>14.7% |
|              | E | 0<br>0.0%         | 2<br>0.0%     | 0<br>0.0%      | 0<br>0.0%      | 643<br>12.2%  | 4<br>0.1%      | 99.1%<br>0.9%  |
|              | F | 75<br>1.4%        | 11<br>0.2%    | 98<br>1.9%     | 72<br>1.4%     | 1<br>0.0%     | 1370<br>26.0%  | 84.2%<br>15.8% |
|              |   | 71.2%<br>28.8%    | 98.8%<br>1.2% | 80.0%<br>20.0% | 86.1%<br>13.9% | 99.7%<br>0.3% | 85.0%<br>15.0% | 89.3%<br>10.7% |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.08$ |               |                |                |               |                |                |
|              | A | 246<br>4.7%       | 2<br>0.0%     | 1<br>0.0%      | 14<br>0.3%     | 0<br>0.0%     | 57<br>1.1%     | 76.9%<br>23.1% |
|              | B | 7<br>0.1%         | 1452<br>27.5% | 4<br>0.1%      | 6<br>0.1%      | 1<br>0.0%     | 32<br>0.6%     | 96.7%<br>3.3%  |
|              | C | 7<br>0.1%         | 0<br>0.0%     | 426<br>8.1%    | 1<br>0.0%      | 0<br>0.0%     | 78<br>1.5%     | 83.2%<br>16.8% |
|              | D | 26<br>0.5%        | 4<br>0.1%     | 0<br>0.0%      | 581<br>11.0%   | 0<br>0.0%     | 93<br>1.8%     | 82.5%<br>17.5% |
|              | E | 0<br>0.0%         | 2<br>0.0%     | 0<br>0.0%      | 0<br>0.0%      | 643<br>12.2%  | 4<br>0.1%      | 99.1%<br>0.9%  |
|              | F | 72<br>1.4%        | 12<br>0.2%    | 93<br>1.8%     | 58<br>1.1%     | 1<br>0.0%     | 1348<br>25.6%  | 85.1%<br>14.9% |
|              |   | 68.7%<br>31.3%    | 98.6%<br>1.4% | 81.3%<br>18.7% | 88.0%<br>12.0% | 99.7%<br>0.3% | 83.6%<br>16.4% | 89.1%<br>10.9% |
|              |   | A                 | B             | C              | D              | E             | F              |                |
|              |   | $\sigma^2 = 0.10$ |               |                |                |               |                |                |
|              | A | 241<br>4.6%       | 5<br>0.1%     | 2<br>0.0%      | 14<br>0.3%     | 0<br>0.0%     | 51<br>1.0%     | 77.0%<br>23.0% |
|              | B | 7<br>0.1%         | 1447<br>27.5% | 4<br>0.1%      | 5<br>0.1%      | 0<br>0.0%     | 26<br>0.5%     | 97.1%<br>2.9%  |
|              | C | 3<br>0.1%         | 0<br>0.0%     | 414<br>7.9%    | 2<br>0.0%      | 0<br>0.0%     | 72<br>1.4%     | 84.3%<br>15.7% |
|              | D | 23<br>0.4%        | 4<br>0.1%     | 0<br>0.0%      | 577<br>10.9%   | 0<br>0.0%     | 80<br>1.5%     | 84.2%<br>15.8% |
|              | E | 0<br>0.0%         | 1<br>0.0%     | 0<br>0.0%      | 0<br>0.0%      | 643<br>12.2%  | 4<br>0.1%      | 99.2%<br>0.8%  |
|              | F | 84<br>1.6%        | 15<br>0.3%    | 103<br>2.0%    | 62<br>1.2%     | 1<br>0.0%     | 1379<br>26.2%  | 83.9%<br>16.1% |
|              |   | 67.3%<br>32.7%    | 98.3%<br>1.7% | 79.0%<br>21.0% | 87.4%<br>12.6% | 99.7%<br>0.3% | 85.5%<br>14.5% | 89.2%<br>10.8% |
|              |   | A                 | B             | C              | D              | E             | F              |                |

Fig. 5. Performance of the proposed method with NVE at different speckle noise levels (right column: precision, bottom row: recall) for the following classes: A: Fetal Abdomen, B: Fetal Brain, C: Fetal Femur, D: Fetal Thorax, E: Maternal Cervix, F: Other.

### 3.2. Feature extraction

A pre-trained CNN architecture  $f_\theta$  is employed in this section. Where  $\theta$  represents the parameters learned during training on the IMAGENET [23] database. This network can be expressed as a composition of layers, transforming the input image  $x_{n,i}$  into a feature representation  $h_i$ :

$$h_i = f_\theta(x_{n,i}) \quad (2)$$

Here,  $f_\theta$  denotes the CNN model with parameters  $\theta$ , which consists of multiple layers including convolutional layers, pooling layers, and fully connected layers. The parameters  $\theta$  of the CNN model are learned during training on the IMAGENET [23] database, where the network is optimized to classify a wide range of natural images into various categories. As a result, the learned parameters  $\theta$  capture rich hierarchical features from the input images, enabling the CNN model to extract meaningful representations of visual content.

In the proposed method, MobileNetV2 [25], ShuffleNet [26], and SqueezeNet [27] are chosen as the pre-trained CNN architectures for feature extraction. These architectures were selected due to their efficiency in terms of model size and computational complexity, making them suitable for environments with limited disk space and resources. MobileNetV2 [25], ShuffleNet [26], and SqueezeNet [27] are designed with fewer parameters compared to traditional CNN architectures, which helps reduce the storage requirements while maintaining competitive performance.

### 3.3. Noise variance estimation

In this section, a method for estimating speckle noise variance in fetal ultrasound images is proposed. The presented noise variance

estimation (NVE) approach utilizes statistical and transform-based features extracted from maternal fetal ultrasound images corrupted with varying levels of speckle noise. A multi-layer perceptron (MLP) is employed to classify noise variance levels into six predefined classes: 0, 0.02, 0.04, 0.06, 0.08, and 0.1. For each input fetal ultrasound image, various statistical and transform-based features are extracted to characterize the noise. These features include singular values from Singular Value Decomposition (SVD), local variance features, wavelet-based features, gradient-based features, high-frequency statistical features from wavelet detail coefficients, low-frequency statistical features from approximation coefficients, and Wiener residual-based features.

The input fetal ultrasound image, denoted as  $x_{n,i}$ , is decomposed using SVD [28]:

$$x_{n,i} = U S V^T \quad (3)$$

where  $U$  and  $V$  are orthogonal matrices, and  $S$  is a diagonal matrix containing singular values. From this decomposition, three key features are extracted: the singular value vector, singular value entropy, and singular value log energy. The singular value entropy is computed as:

$$H_s = - \sum s_i \log(s_i + \epsilon) \quad (4)$$

and the singular value log energy is computed as:

$$E_s = \sum \log(s_i + \epsilon) \quad (5)$$

where  $s_i$  are the singular values, and  $\epsilon$  is a small constant to prevent numerical instability.

Local variance measures how pixel intensities vary within a small neighborhood around each pixel [29]. Here, local variance is computed using a  $5 \times 5$  sliding window. The mean and standard deviation of the

local variance values are extracted as features:

$$\mu_{LV} = \frac{1}{N} \sum_{i=1}^N LV_i \quad (6)$$

$$\sigma_{LV} = \sqrt{\frac{1}{N} \sum_{i=1}^N (LV_i - \mu_{LV})^2} \quad (7)$$

where  $LV_i$  represents the local variance values, and  $N$  is the number of pixels in the image.

The discrete wavelet transform (DWT) using the Haar wavelet decomposes the image into approximation ( $cA$ ) and detail coefficients ( $cH, cV, cD$ ) [30]. The wavelet energy is computed as:

$$E_w = \frac{1}{N} \sum (|cH| + |cV| + |cD|) \quad (8)$$

The high-frequency components ( $cH, cV, cD$ ) provide four statistical features: mean ( $\mu_H$ ), standard deviation ( $\sigma_H$ ), skewness ( $Sk_H$ ), and kurtosis ( $Ku_H$ ). Similarly, the approximation component ( $cA$ ) provides four analogous features: mean ( $\mu_L$ ), standard deviation ( $\sigma_L$ ), skewness ( $Sk_L$ ), and kurtosis ( $Ku_L$ ).

Adaptive Wiener filtering [31] is applied to obtain a residual image:

$$R_W = x_{n,i} - W(x_{n,i}) \quad (9)$$

where  $W(x_{n,i})$  is the Wiener-filtered image. The standard deviation of this residual,  $\sigma_W$ , is used as a feature. In addition, image gradients [32] are analyzed to capture edge information, extracting the mean and standard deviation of gradient magnitudes, denoted as  $\mu_{grad}$  and  $\sigma_{grad}$ , respectively. A fully connected MLP is designed for noise variance classification. The network consists of an input layer that receives the extracted feature vector, followed by two hidden layers with 128 and 64 neurons, each employing the ReLU activation function. The output layer consists of six neurons with a softmax activation function, corresponding to the six noise variance levels.

### 3.4. Classification

The feature vector  $h_i$  extracted from the maternal–fetal ultrasound plane is given to SVM to facilitate classification. Mathematically, the SVM training process involves finding the hyperplane that maximizes the margin between different classes while minimizing classification error. During the training process, the SVM algorithm optimizes the parameters ( $\alpha_i$  and  $b$ ) to find the hyperplane that best separates the classes in the feature space. After training, the SVM model can classify new maternal–fetal ultrasound image planes by computing the decision function for each image. The SVM classifier with a radial basis function kernel and one-vs-all strategy is implemented in this work for multi-class classification of maternal–fetal ultrasound planes.

In this study, distinct SVM classifiers are trained individually for various maternal–fetal ultrasound images at different speckle noise variances ranging from 0 to 0.1 with step size of 0.02. Following training, these SVM classifiers are stacked, as depicted in Fig. 3, and organized within a switch control framework. Notably, it becomes apparent that the performance of the proposed method is consistent when features extracted from the input maternal–fetal ultrasound images are subsequently fed to the specific SVM trained for the variance equal to the variance of the noise in the input fetal ultrasound image.

### 3.5. Switch control

Let  $\sigma^2$  represent the noise variance of the input ultrasound image. In this study, a switch control mechanism is introduced, where the switch, denoted as  $S_{\sigma^2}$ , is operated based on the noise variance. When the noise variance is  $\sigma^2$ , the switch  $S_{\sigma^2}$  is closed to map the input maternal–fetal ultrasound plane to a SVM model trained for variance  $\sigma^2$ .

**Table 1**

Summary of the dataset FETAL\_PLANES\_DB including the distribution of images.

| Class           | Training images | Testing images |
|-----------------|-----------------|----------------|
| Fetal abdominal | 353             | 358            |
| Fetal brain     | 1620            | 1472           |
| Fetal femur     | 516             | 524            |
| Fetal thorax    | 1058            | 660            |
| Maternal cervix | 981             | 645            |
| Others          | 2601            | 1612           |
| <b>Total</b>    | <b>7129</b>     | <b>5271</b>    |

Mathematically, the switch control mechanism can be expressed as follows:

$$S_{\sigma^2} = \begin{cases} 1 & \text{if } \sigma^2 = \sigma_{\text{target}}^2 \\ 0 & \text{otherwise} \end{cases}$$

where  $\sigma_{\text{target}}^2$  is the target noise variance for which the SVM model is trained. When the switch  $S_{\sigma^2}$  is closed (i.e.,  $S_{\sigma^2} = 1$ ), the input image  $x_i$  is mapped to the SVM model trained for noise variance  $\sigma^2$ . The switch control mechanism enables adaptive classification based on the noise characteristics of the input fetal ultrasound image.

### 3.6. Evaluation

The evaluation metrics such as diagnostic accuracy, precision, and recall are used in this work to evaluate the effectiveness of the proposed method. These evaluation metrics play crucial roles in assessing the effectiveness of a classification model, including in applications such as prenatal care, where the accurate identification of ultrasound imaging planes is essential.

## 4. Result analysis

### 4.1. Dataset characteristics

In this work, the maternal–fetal ultrasound image planes are obtained from the FETAL\_PLANES\_DB: Common maternal–fetal ultrasound images dataset [5]. The dataset [5] includes fetal ultrasound images, comprising a total of 12,400 images in PNG format from 1792 patients, devoid of any patient identifying information to ensure ethical compliance. Each image represents various aspects of fetal anatomy. The dataset was collected at BCNatal, a center with two sites: Hospital Clinic and Hospital Sant Joan de Déu in Barcelona, Spain. The fetal ultrasound images are organized by patient and study date, with the first half (patient numbers 1 to 1226) used for training and the second half (patient numbers 1227 to 1792) for testing. The dataset includes ultrasound images from patients in the second and third trimesters. The imaging was performed using Voluson E6, Voluson S8, Voluson S10, and Aloka machines, with frequency ranges of 3 to 7.5 MHz for abdominal probes and 2 to 10 MHz for vaginal probes. Detailed information about the dataset is provided in Table 1.

### 4.2. Experimental setup

In this study, experiments are conducted using MATLAB 2024a equipped with the image processing, deep learning, and statistics and machine learning toolboxes. The hardware configuration used in this study as follows:

- Processor: 11th Gen Intel(R) Core(TM) i7-11700F @ 2.50 GHz
- RAM: 16 GB
- Graphics Card: NVIDIA GeForce GTX 1650

**Table 2**

Performance of the proposed ensemble method against different pre-trained CNNs.

| Pre-trained net      | Diagnostic accuracy (in %) at different speckle noise levels |                   |                   |                   |                   |                  |
|----------------------|--------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|------------------|
|                      | $\sigma^2 = 0$                                               | $\sigma^2 = 0.02$ | $\sigma^2 = 0.04$ | $\sigma^2 = 0.06$ | $\sigma^2 = 0.08$ | $\sigma^2 = 0.1$ |
| SqueezeNet [27]      | 87.27                                                        | 76.63             | 68.09             | 61.43             | 56.25             | 51.24            |
| GoogLeNet [33]       | 87.67                                                        | 80.90             | 74.14             | 68.07             | 63.61             | 59.10            |
| InceptionV3 [34]     | 87.82                                                        | 80.31             | 77.39             | 76.40             | 75.03             | 73.99            |
| DenseNet-201 [35]    | 89.00                                                        | 79.51             | 70.08             | 63.93             | 59.70             | 55.55            |
| MobileNetV2 [25]     | 87.97                                                        | 79.98             | 76.44             | 73.59             | 71.35             | 68.60            |
| AlexNet [36]         | 89.21                                                        | 85.01             | 78.45             | 71.09             | 64.52             | 58.64            |
| ResNet-18 [37]       | 86.11                                                        | 78.73             | 74.46             | 71.73             | 68.70             | 65.91            |
| ResNet-50 [37]       | 88.31                                                        | 78.14             | 68.96             | 62.13             | 56.54             | 52.74            |
| ResNet-101 [37]      | 88.09                                                        | 77.92             | 72.42             | 67.65             | 63.61             | 60.42            |
| Xception [38]        | 86.28                                                        | 80.76             | 77.48             | 74.63             | 73.48             | 70.86            |
| ShuffleNet [26]      | 88.84                                                        | 70.84             | 64.37             | 58.34             | 54.03             | 49.31            |
| DarkNet-19 [39]      | 86.19                                                        | 71.41             | 61.39             | 54.92             | 49.90             | 46.82            |
| DarkNet-53 [39]      | 88.29                                                        | 71.94             | 62.21             | 56.31             | 52.55             | 49.38            |
| EfficientNetB0 [40]  | 88.07                                                        | 81.45             | 77.12             | 73.86             | 70.78             | 68.32            |
| VGG-16 [41]          | 86.38                                                        | 82.07             | 79.11             | 76.30             | 73.65             | 70.80            |
| VGG-19 [41]          | 86.06                                                        | 83.95             | 80.54             | 76.40             | 72.89             | 69.34            |
| Proposed without NVE | <b>90.17</b>                                                 | <b>89.83</b>      | <b>89.39</b>      | <b>89.34</b>      | <b>89.09</b>      | <b>89.19</b>     |
| Proposed with NVE    | <b>90.17</b>                                                 | <b>89.83</b>      | <b>89.39</b>      | <b>89.34</b>      | <b>89.09</b>      | <b>89.19</b>     |

#### 4.3. Performance of the proposed method

The proposed method introduces an innovative ensemble approach by combining multiple pre-trained CNNs for maternal–fetal ultrasound image classification. Specifically, the method explores the performance of MobileNetV2 [25], ShuffleNet [26], and SqueezeNet [27]. The performance of this ensemble is extensively evaluated using the FETAL\_PLANES\_DB [5] dataset at different speckle noise levels. As shown in Table 2, the MobileNetV2 [25], ShuffleNet [26], and SqueezeNet [27] ensemble outperforms other pre-trained CNN models, demonstrating superior classification accuracy. The results in Table 2 highlight that the proposed ensemble method with and without NVE provides better performance compared to state-of-the-art pre-trained models present in the literature. Additionally, the confusion matrices of the proposed method with and without NVE at various speckle noise levels are presented in Figs. 4 and 5.

From Table 2, it is observed that the proposed method demonstrates similar performance with and without the NVE module in the current experimental setup. This suggests that the pretrained feature extractor is already capturing robust representations, making the impact of noise variance estimation less pronounced in this scenario. However, the NVE module remains essential in cases where higher noise levels significantly degrade feature quality, as it enables adaptive classifier selection for improved robustness. The extracted features are given to dedicated SVM classifiers, and a majority voting scheme is applied to aggregate their predictions, contributing to the overall improved performance, as shown in Table 2.

#### 4.4. Impact of switch control using noise variance estimation

As depicted in Fig. 6, the performance of the proposed method is extensively evaluated across different speckle noise levels using the FETAL\_PLANES\_DB [5] dataset. The individual pre-trained networks used in the proposed method, namely MobileNetV2 [25], ShuffleNet [26], and SqueezeNet [27], along with the ensemble of these networks are highlighted in Fig. 6. During the evaluation, it is observed that when SVMs trained at high noise levels are tested on low noise levels, the performance is significantly reduced. Similarly, SVMs trained at low noise levels do not perform well when evaluated on images with high noise levels. This observation points to a clear inconsistency in performance when the SVM models are not matched with the appropriate noise characteristics of the input fetal ultrasound data.

In the context of noise, low speckle noise levels correspond to a variance of 0, while high noise speckle levels are associated with a variance of 0.1. This discrepancy in performance can be attributed to

the fact that the parameters of the CNN model are influenced by the noise conditions during training, leading to suboptimal results when there is a mismatch between the noise levels of the training and test data. To address this challenge, it is critical to perform noise variance mapping, ensuring that the correct SVM model is selected based on the variance of the test input ultrasound plane.

The proposed method incorporates a noise variance mapping mechanism along with a switch control strategy that adapts to different noise levels, ensuring consistent performance across varying levels of speckle noise. The ensemble of MobileNetV2 [25], ShuffleNet [26], and SqueezeNet [27] uses the strengths of each individual model to enhance the robustness of the classification task, particularly in the presence of noise. The switch control mechanism is key to ensure that the ensemble can dynamically adjust to speckle noise conditions, further improving classification accuracy and reliability.

In this work, NVE approach is proposed using statistical and transform-based features. The proposed NVE is trained and tested on the FETAL\_PLANES\_DB dataset [5], as detailed in Table 1. The network is optimized using Stochastic Gradient Descent with Momentum. The training parameters include a learning rate of 0.0001, a mini-batch size of 64, and a maximum of 400 epochs, with validation performed every 100 iterations. Training progress is monitored using accuracy and loss metrics. After training, the model is evaluated on the test dataset, where its performance is assessed using classification accuracy and a confusion matrix, as illustrated in Fig. 7. The proposed method achieves a classification accuracy of 93.8% in predicting the correct noise variance level. Ultimately, the approach demonstrates consistent performance in effectively estimating noise variance and selecting the appropriate SVM model, proving its robustness in handling real-world variations in ultrasound image quality.

#### 4.5. Comparison with existing methods

A detailed comparison of diagnostic accuracy achieved by various methods [5,7,12,13,15,20–22] at different speckle noise levels (denoted by  $\sigma^2$  values) is provided in Table 3. Among the methods, Chen et al. [12] and Burgos et al. [5] demonstrate strong performance at lower noise levels ( $\sigma^2 = 0$ ), with accuracies of 91.50% and 92.80%, respectively. However, their performance degrades significantly as noise levels increase, particularly for Chen et al. [12], which drops to 69.70% at  $\sigma^2 = 0.1$ . Similarly, Baumgartner et al. [13] and Yu et al. [15] show drastic accuracy declines at higher noise levels, indicating their limited robustness to speckle noise. Conversely, Krishna et al. [20–22] exhibit better stability across noise levels, maintaining accuracies above 80% at  $\sigma^2 = 0.1$ .

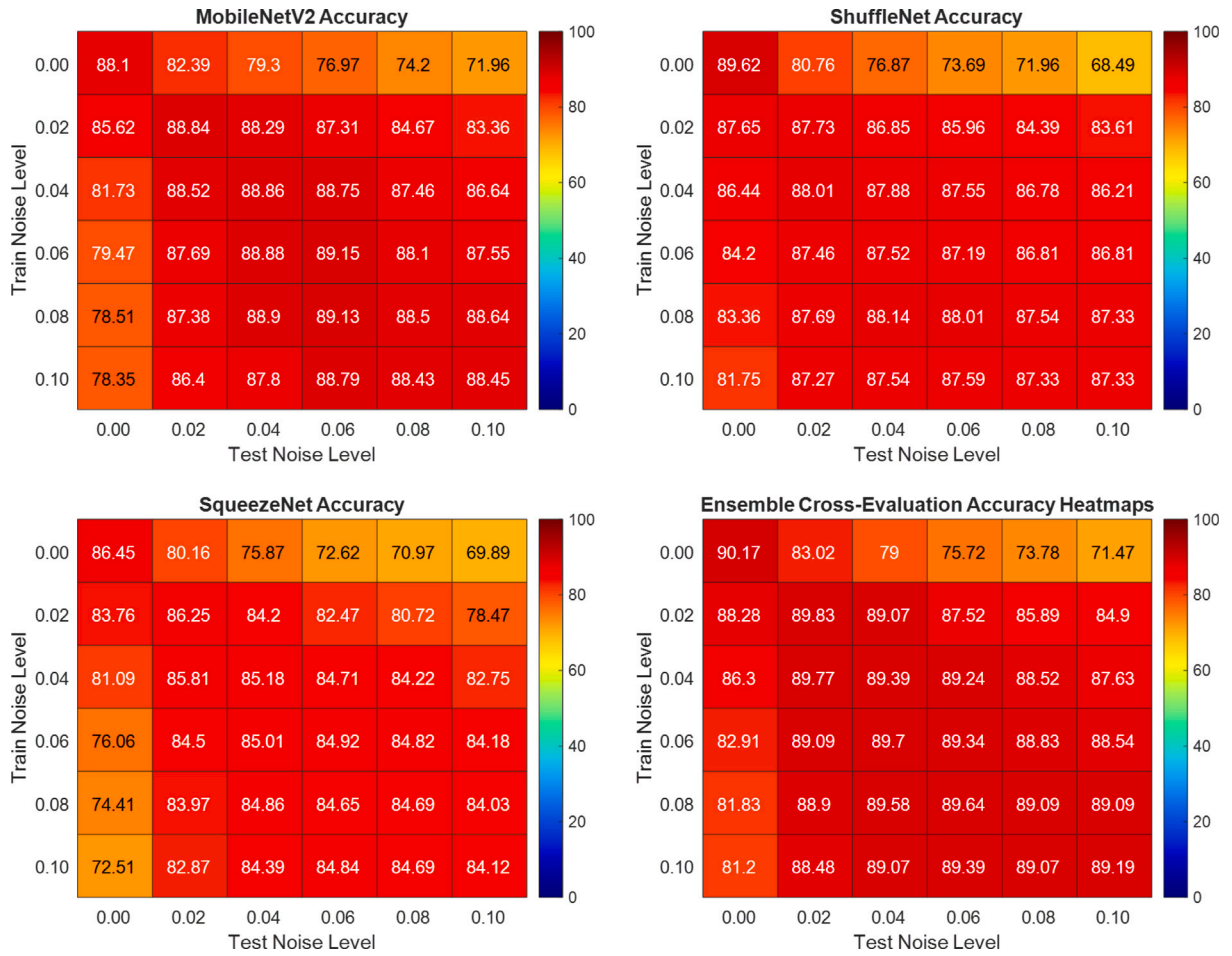


Fig. 6. Impact of switch control based on noise variance estimation for SVM Selection and Consistent Performance.

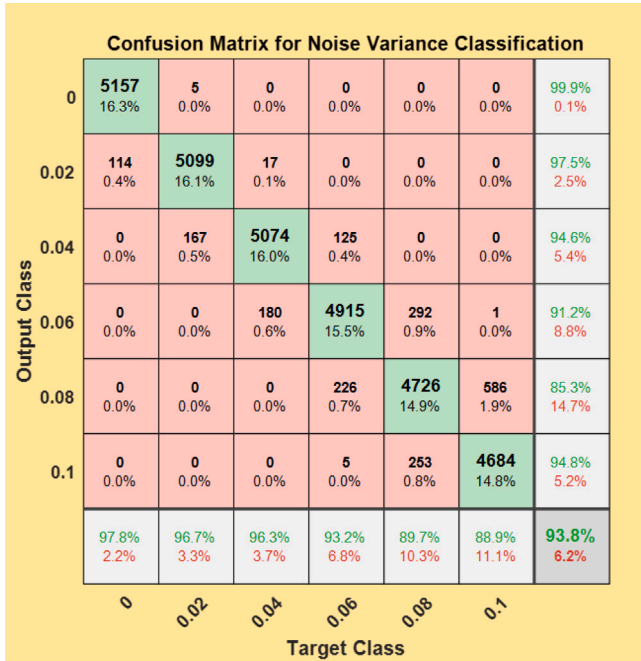


Fig. 7. Confusion matrix for the proposed NVE approach (right column: precision, bottom row: recall).

The proposed method stands out due to its consistent performance across all noise levels. Unlike existing methods [5,7,12,13,15,20–22], which experience significant accuracy drops, the proposed approach maintains stable diagnostic accuracy, ranging from 90.17% at  $\sigma^2 = 0$  to 89.19% at  $\sigma^2 = 0.1$ . This consistent performance highlights the robustness of the proposed method in handling varying noise levels, attributed to its adaptive noise mapping and switch control mechanism, as well as the ensemble of MobileNetV2 [25], ShuffleNet [26], and SqueezeNet [27]. The proposed method not only provides competitive results under ideal conditions but also outperforms state-of-the-art methods in terms of robustness under noisy conditions.

The performance of the proposed method and existing methods [5,7,12,13,15,20–22] are illustrated using confusion matrices, are presented in Fig. 8. In these confusion matrices, the rightmost column represents precision, while the bottommost row represents recall. From Fig. 8, it is evident that the diagnostic accuracy of the proposed method is on par with other existing methods. All the existing methods [5,7,12,13,15,20–22], including the proposed approach, are evaluated using the FETAL\_PLANES\_DB [5] dataset, ensuring that training and testing is conducted on entirely different and unseen patient data. To maintain consistency, the existing methods were re-evaluated under the same conditions as the proposed method, with their parameter settings kept unchanged.

#### 4.6. Computational complexity

A comprehensive comparison of various methods for fetal ultrasound image classification, focusing on key factors such as the type of learning, disk size of the CNN, fine-tuning time, and diagnostic

Table 3

Comparison of the proposed method with existing methods in fetal ultrasound plane classification.

| Method   | Diagnostic accuracy (in %) at different speckle noise levels |                   |                   |                   |                   |                  |
|----------|--------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|------------------|
|          | $\sigma^2 = 0$                                               | $\sigma^2 = 0.02$ | $\sigma^2 = 0.04$ | $\sigma^2 = 0.06$ | $\sigma^2 = 0.08$ | $\sigma^2 = 0.1$ |
| [12]     | 91.50                                                        | 89.50             | 84.50             | 78.80             | 74.30             | 69.70            |
| [13]     | 90.30                                                        | 45.20             | 32.30             | 30.70             | 30.60             | 30.60            |
| [5]      | 92.80                                                        | <b>92.00</b>      | 90.50             | 88.50             | 86.20             | 83.60            |
| [7]      | 92.20                                                        | 89.60             | 86.00             | 83.80             | 82.10             | 80.30            |
| [15]     | 90.30                                                        | 60.20             | 38.40             | 32.30             | 31.00             | 30.70            |
| [20]     | 91.90                                                        | 90.10             | 87.20             | 84.90             | 82.50             | 80.40            |
| [21]     | 92.70                                                        | 91.90             | <b>90.90</b>      | 89.30             | 87.00             | 84.80            |
| [22]     | <b>93.00</b>                                                 | 90.90             | 88.70             | 85.90             | 83.10             | 80.50            |
| Proposed | 90.17                                                        | 89.83             | 89.39             | <b>89.34</b>      | <b>89.09</b>      | <b>89.19</b>     |

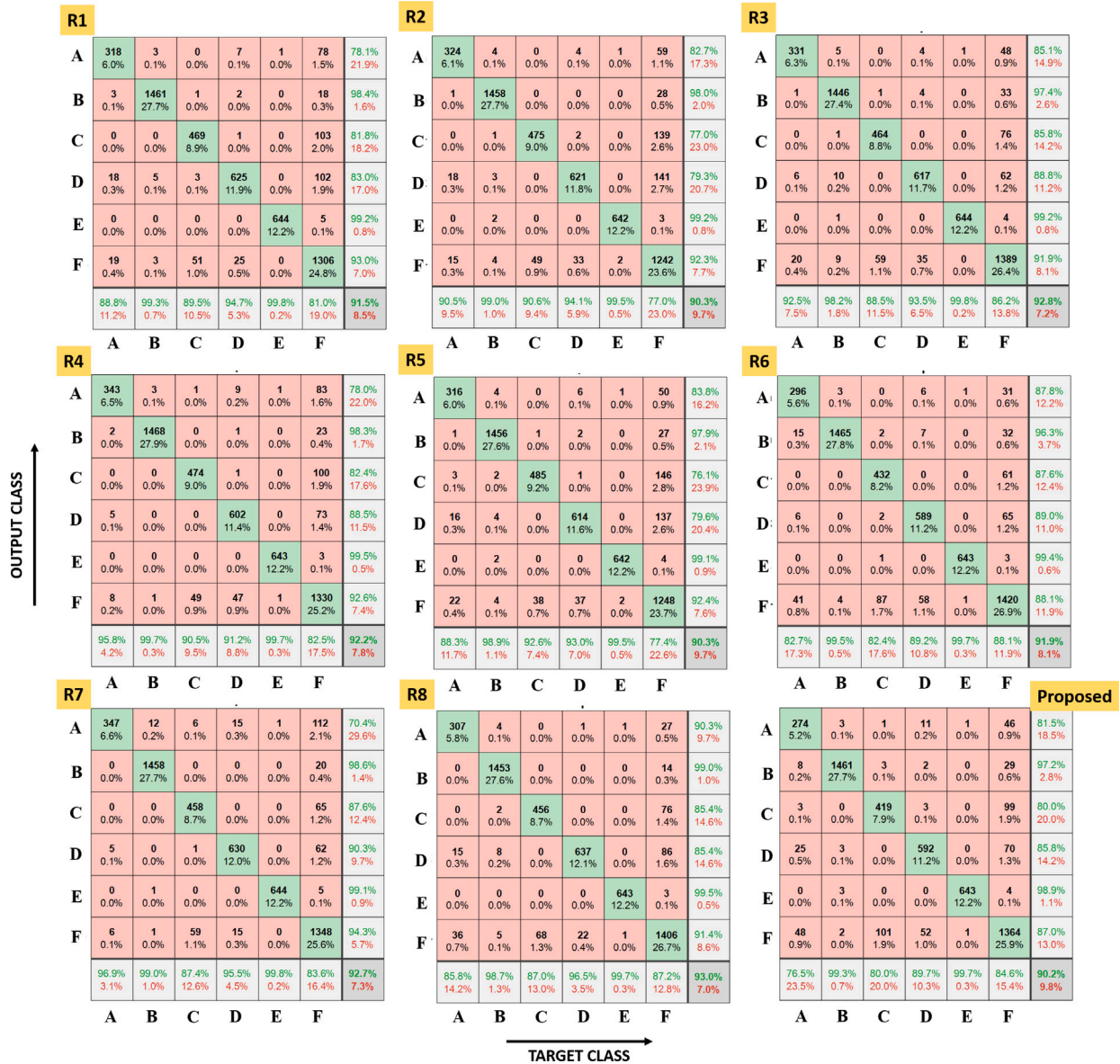


Fig. 8. Comparison of diagnostic accuracy across various methods (R1: [12], R2: [13], R3: [5], R4: [7], R5: [15], R6: [20], R7: [21], R8: [22]) and the proposed method (right column: precision, bottom row: recall) for the following classes: A: Fetal Abdomen, B: Fetal Brain, C: Fetal Femur, D: Fetal Thorax, E: Maternal Cervix, F: Other.

accuracy, is presented in Table 4. The existing methods [5,7,12,13,15,20–22] listed in Table 4 utilize fine-tuning of pre-trained models. While fine-tuning typically results in high diagnostic accuracy, there are significant variations in both the disk size of the CNN models and the time required for fine-tuning. For instance, Baumgartner et al. [13] achieved an accuracy of 90.3% with a model size of 52.8 MB, but it

took 755 min for fine-tuning. Ghabri et al. [7] achieved slightly higher accuracy (92.2%) with a model size of 83.7 MB and a much shorter fine-tuning time of 15 min. The proposed method utilizes pre-trained models instead of fine-tuning, offering a distinct advantage in terms of computational efficiency. It requires only 22.14 MB of disk space and eliminates the need for fine-tuning (with a time of 0 s). Despite these

**Table 4**  
Comparison of computational complexity in terms of fine-tuning time and disk size.

| Method          | Type of learning | Disk size of CNN (in MB) | Time to finetune CNN (in min) | Diagnostic accuracy (%) |
|-----------------|------------------|--------------------------|-------------------------------|-------------------------|
| [12]            | Fine-tuning      | 217.00                   | 11                            | 91.50                   |
| [13]            | Fine-tuning      | 52.80                    | 755                           | 90.30                   |
| [5]             | Fine-tuning      | 78.30                    | 472                           | 92.80                   |
| [7]             | Fine-tuning      | 83.70                    | 15                            | 92.20                   |
| [15]            | Fine-tuning      | 54.20                    | 753                           | 90.30                   |
| [20]            | Fine-tuning      | 277.80                   | 534                           | 91.90                   |
| [21]            | Fine-tuning      | 84.20                    | 559                           | 92.70                   |
| [22]            | Fine-tuning      | 93.40                    | 611                           | 93.00                   |
| <b>Proposed</b> | Pre-trained      | <b>22.29</b>             | <b>0</b>                      | 90.17                   |

computational benefits, the proposed method achieves a diagnostic accuracy of 90.17%, which is comparable to other fine-tuning-based methods. This highlights the strength of the proposed approach, as it strikes a balance between maintaining competitive classification performance and significantly reducing both the computational cost and time required for model training.

## 5. Discussions

The proposed method for maternal–fetal ultrasound plane is compared with various existing methods [5,7,12,13,15,20–22] focusing on parameters such as learning type, model size, fine-tuning time, and diagnostic accuracy. Existing methods [5,7,12,13,15,20–22] rely heavily on fine-tuning pre-trained CNN models, which often results in higher diagnostic accuracy but comes at the cost of significant computational resources. For example, Baumgartner et al. [13] achieved an accuracy of 90.3% with a model size of 52.8 MB, requiring 755 min for fine-tuning. Similarly, Ghabri et al. [7] achieved a slightly higher accuracy of 92.2% with an 83.7 MB model and a reduced fine-tuning time of 15 s. In contrast, the proposed method eliminates the need for fine-tuning altogether, utilizing pre-trained lightweight models. With a minimal disk size of 22.14 MB and no additional training time, the method achieves a competitive accuracy of 90.17%, underscoring its computational efficiency without compromising performance.

A notable strength of the proposed approach lies in its robust performance across varying noise levels, as shown in Table 3. While methods like Chen et al. [12] and Baumgartner et al. [13] demonstrate significant accuracy drops under increasing speckle noise conditions, the proposed method exhibits remarkable stability. In contrast, the proposed method maintains consistent accuracy, ranging from 90.17% at  $\sigma^2 = 0$  to 89.19% at  $\sigma^2 = 0.1$ . This resilience can be attributed to the ensemble of lightweight CNNs such as MobileNetV2 [25], ShuffleNet [26], and SqueezeNet [27] combined with the adaptive noise variance mapping and switch control mechanism. Using larger datasets of 12,400 images and computationally intensive fine-tuning of models like AlexNet [36] and VGG19 [41], the proposed method achieves comparable performance with significantly reduced data and resource requirements. By exploring pre-trained models to extract generalizable feature representations, the presented method ensures robustness and adaptability, particularly in scenarios with limited data availability. These attributes, coupled with its computational efficiency and resilience to noise, position the proposed method as a viable and practical solution for maternal–fetal ultrasound image classification in diverse clinical and real-world settings.

## 6. Conclusion

In this paper, an approach for accurate classification of maternal–fetal ultrasound images by exploiting the combined power of pre-trained CNNs and SVMs. By employing a pre-trained CNN as a feature extractor, important spatial and structural information relevant to ultrasound imaging is captured from the images. These high-dimensional

feature representations are then fed into an SVM classifier using proposed NVE approach, allowing for robust and effective classification of ultrasound fetal planes. The proposed methodology demonstrates promising results in accurately identifying ultrasound imaging planes, which is crucial for prenatal care applications. Through experimentation and evaluation, it is evident that the integration of pre-trained CNNs and SVMs leads to improved classification performance in terms of computational complexity compared to traditional methods. Future work will focus on developing methods to handle unknown or varying noise levels, further enhancing the robustness of the proposed approach.

## CRedit authorship contribution statement

**Turimerla Pratap:** Software, Methodology, Data curation, Conceptualization. **Venkata Rao Dhulipalla:** Validation, Supervision, Resources, Conceptualization. **Priyanka Kokil:** Writing – original draft, Supervision, Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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